Wholesale Customer Segmentation Using PCA and Clustering: A Comparison with LLM-Based Analysis

(Word Count: 1,024)

1. Introduction

Customer segmentation is widely adopted by firms to efficiently market their products and organize their services (Cooil et al., 2008). With the evolving digital revolution, businesses are increasingly utilizing machine learning algorithms to better understand their customer portfolio (Mozumder et al., 2024; Mufarroha et al., 2022). However, similar to humans, AI can also carry out this task. In this study, our aim is to conduct a customer segmentation analysis and compare our findings with those generated by a large language model.

2. Data Description

The dataset represents 440 customers, including six numerical features capturing annual spending on wholesale products such as Milk, Delicatessen products etc., and two categorical features describing customer characteristics, Channel and Region (see Appendix A, Table A1). Descriptive statistics show that all numerical features are highly right-skewed with outliers, and the categorical variables show imbalance. In addition, several variables are strongly correlated (see Appendix B for EDA).

3. Methodology

Initially, the dataset was standardized for equal feature contribution (Kumar et al., 2025). PCA was applied to reduce dimensionality and handle multicollinearity (Kurita, 2021). The optimal number of components was determined using the elbow method and the cumulative explained variance ratio since additional components can make distance measurements less informative and induce noise (Schmitt et al., 2016; Youness & Mohamed, 2025). These were adopted as inputs in clustering to improve clustering stability (Youness & Mohamed, 2025). Two clustering approaches were selected for the comparison: K-means and hierarchical clustering. For hierarchical clustering, Euclidean distance was used with Complete or Ward linkage. The determination of K and clustering quality was evaluated using the silhouette score and interpretability, since the silhouette score can be insufficient alone (Myagmarsuren, 2024; Shahapure & Nicholas, 2020). Finally, Gemini was used to perform the same tasks, and the results were compared.

4. Principal Component Analysis (PCA)

PCA was applied to summarise the main patterns in customer spending and to reveal purchasing strategies. Three components were selected, explaining 84.79% of the total variance (44.08%, 28.38%, 12.33%). As seen in Table 2, each component captures different dimensions of customer purchasing behaviours. The first principal component (PC1) separates customers who have high spending on local demand products from others. The second principal component (PC2) reflects spending on culinary and food-preparation purchases, with higher scores indicating higher spending. The third principal component (PC3) is related to the distinction between customers whose spending is more oriented towards fresh purchasing from those with a more delicatessen-focused profile. As also supported by Appendix B, Figure B5, the average PCA scores by channel, where Channel 2 scores higher on PC1 and Channel 1 scores higher on PC2. This shows the difference between retail-style purchasing and culinary or food-preparation demand (see Appendix B for PCA results).

Table 2

Principal Component Loadings for the First Three Components

	Fresh	Milk	Grocery	Frozen	Detergents paper	Delicassen
PCA 1	0.0429	0.5451	0.5793	0.0512	0.05486	0.02487
PCA 2	0.5279	0.0832	-0.1460	0.6113	-0.2552	0.5042
PCA 3	0.8123	-0.0604	0.1084	-0.1784	0.1362	-0.5239

Note. Loadings are reported for the retained principal components after standardising the variables. Larger absolute values indicate a stronger contribution of the variable to the component, while positive and negative signs indicate the direction of the relationship.

5. Clustering

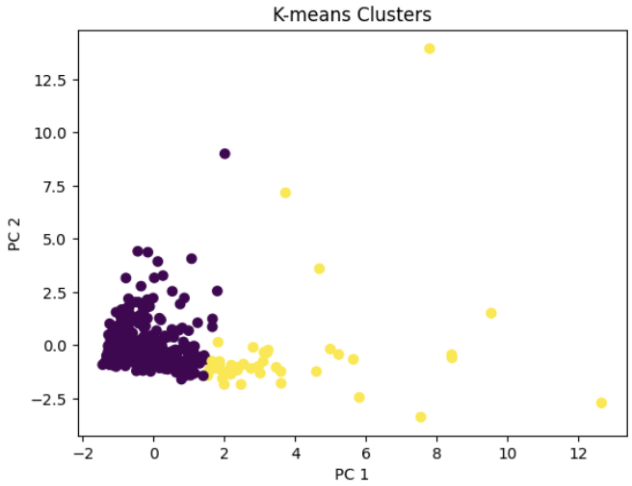
Clustering analysis using various approaches suggested that the adopted approach results in different clusters in terms of their meaning and quality. For K-means clustering, the solution was selected as two clusters (K=2), with the highest overall silhouette score (0.64) among the alternatives. As Figure 1 displays, Cluster 0 contained the majority of the observations (395). It further suggests that the main separation between Clusters is primarily along PC1, where Cluster 1 shows more variation along the PC1 axis. This is

further supported by Appendix D.1, Figure D.1.1, where Cluster 1 is positioned at substantially higher positive values. Thus, these suggest that Cluster 0 represents low domestic consumption customers, while Cluster 1 represents high domestic consumption customers.

Furthermore, the hierarchical clustering results obtained using Ward and complete linkage reveal notable differences in cluster balance and interpretability. Ward linkage identifies three clusters with sizes of 415, 16, and 9, with a slightly lower silhouette score than complete linkage (0.68). Meanwhile, complete linkage produces more imbalanced clusters (420,15,4,1) with a notable singleton cluster which represents an extreme outlier. While clusters under both linkages vary along with PC1, Ward linkage also accounts for PC2 to distinguish between small clusters (see Figure 2 and see Appendix D.2, Figure D.2.3 for more detail). Moreover, Table 3 shows that Cluster 1 displays a specialized consumption pattern, while Cluster 2 shows a retail channel domination. Thus, Ward generates three interpretable clusters: mainstream low domestic consumption customers, a niche segment of culinary-oriented consumers in the horeca channel, and a small group of retail-oriented high domestic consumption customers. These clusters are more meaningful and interpretable compared to those of complete linkage.

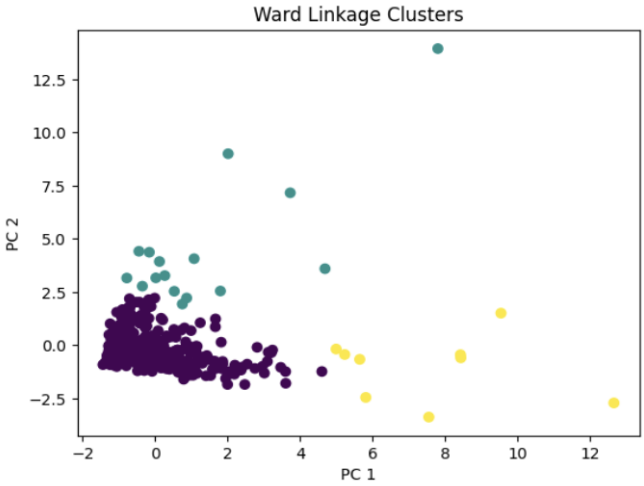
Choosing the best model between K-means and Ward hierarchical clustering depends on the business goal. Since K-means yields a simple two-cluster solution driven by domestic consumption, it may be a particularly useful tool for operational use, such as assigning customers to available salespersons. Conversely, Ward clustering, with its more detailed segments, is more appropriate for strategic market decisions such as product positioning and differentiated value propositions.

Figure 1
K-means Clusters in PCA-Transformed Feature Space



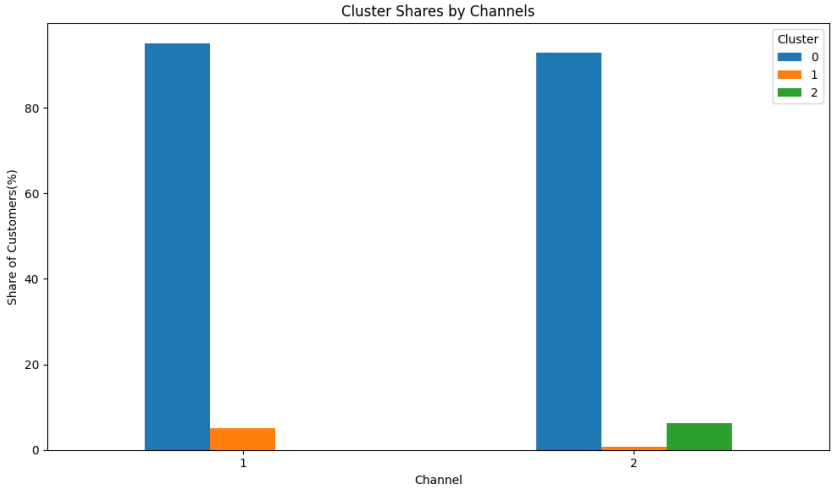
Note: The majority of observations are in Cluster 0, purple. Clusters mainly differ across the PC1 axis. The main difference between clusters is the domestic consumption behaviour.

Figure 2
Ward Linkage Clusters in PCA-Transformed Feature Space



Note: The majority of observations are in Cluster 0, purple. Clusters show variation across both PC1 and PC2. Small clusters are differentiated by PC2.

Figure 3
Distribution of Customer Clusters Across Channels



Note: Cluster 0 dominates both Channel 1 (Horeca) and Channel 2 (Retail), which indicates that, regardless of the channel, most customers belong to the mainstream cluster. Cluster 1 appears only marginally in both channels but more in Channel 1, indicating a specialised consumption pattern rather than channel domination, while Cluster 2 shows a retail channel domination.

6. Large Language Model (LLM)

The LLM performed the same PCA and clustering tasks using K-Means and Hierarchical Clustering (see Appendix E for prompts). Overall, it followed a similar pipeline, including standardising before PCA and retaining two principal components for clustering. Although it adopted similar steps, its approaches to interpretation and model selection differed. The results are generated quickly, however, LLM provides insufficient for interpretation and explanation. A key limitation is that output quality is highly prompt-dependent (Machlab & Battle, 2024). Further prompting was needed to get clearer justification and deeper insights, which suggests that the initial outputs were limited in depth. For example, for K-Means clustering, while LLM selected three clusters based only on the elbow method, we also considered silhouette scores. The LLM only considered this after prompting. The LLM only used Ward linkage for the Hierarchical Clustering and chose three clusters. After further prompting, it gave some justification and limited interpretation. Overall, the LLM can efficiently execute the standard pipeline steps but requires substantial human input to produce well-justified, interpretable results.

7. Conclusion

This study showed that customer segmentation can generate meaningful business insights when some statistical methods are carefully combined. PCA revealed three main purchasing dimensions that present the strategies behind the customer's behaviours. Clustering showed that the most useful solution depends on the business goal. K-means offered simple operational use while Hierarchical clustering is a useful tool for marketing decisions. And finally, LLMs can support customer segmentation tasks, but to achieve reliable results, it still needs human judgment.

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Appendices

Appendix A: Dataset Overview

Table 1

Dataset Variables and Descriptions

Variable	Description
Channel	Customer channel: Horeca (Hotel/Restaurant/Café) or Retail
Region	Customer region: Lisbon, Oporto, or Other
Fresh	Annual spending (m.u.) on fresh products
Milk	Annual spending (m.u.) on milk products
Grocery	Annual spending (m.u.) on grocery products
Frozen	Annual spending (m.u.) on frozen products
Detergents_Paper	Annual spending (m.u.) on detergents and paper products
Delicatessen	Annual spending (m.u.) on delicatessen products

Note. The table shows the variables and their descriptions in the wholesale customer dataset, including annual spending across six product categories and customer classification by channel and region.

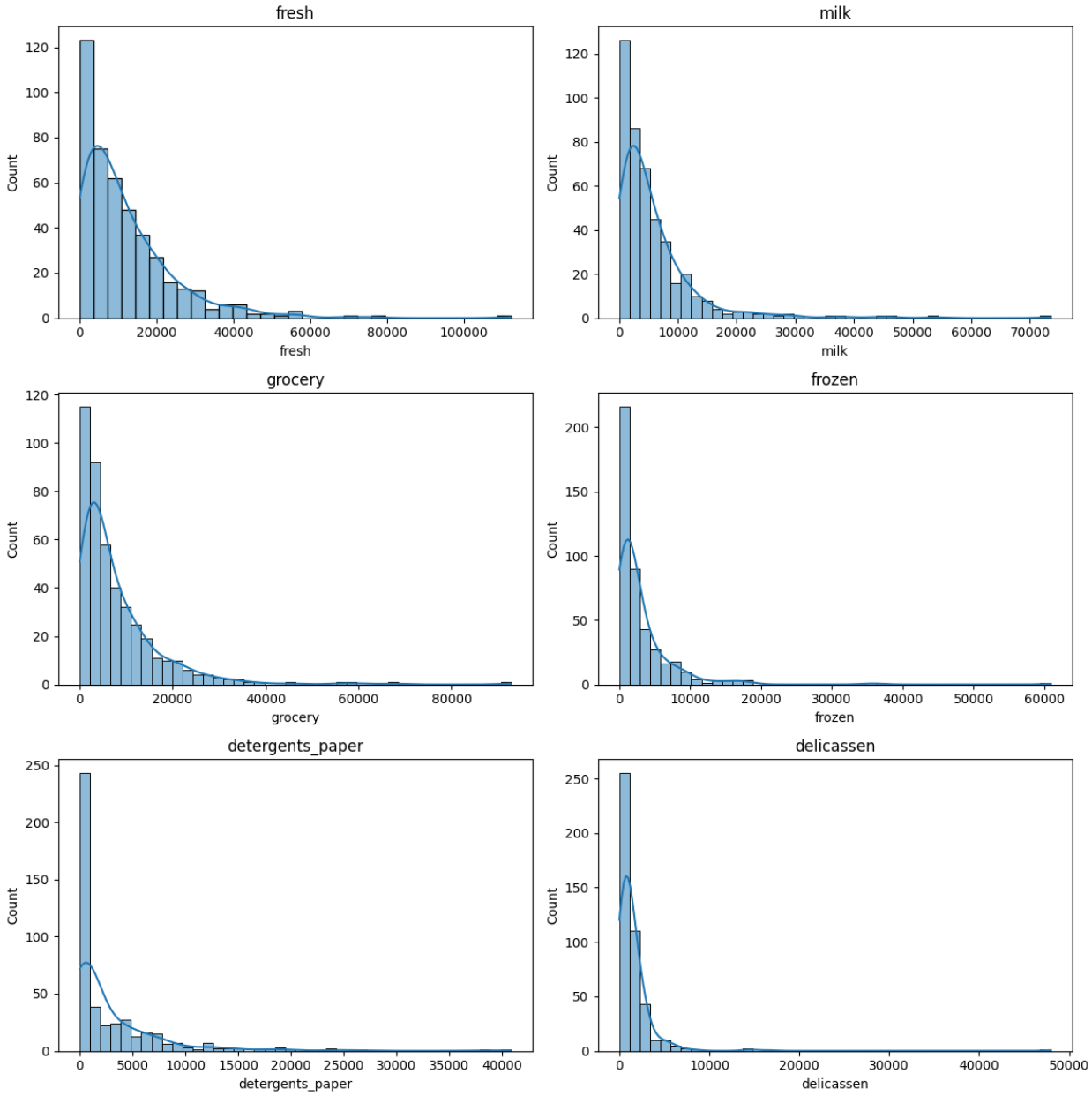
Appendix B: Exploratory Data Analysis (EDA)

Exploratory data analysis shows several important patterns in the dataset. The histograms in Figure B.1 reveal how all six variables are strongly right skewed. This means that most of the customers spend relatively low amounts while a smaller group spends substantially more. These results can also be seen from Figure B.2 which shows the extreme values across almost all products. The categorical variables also display uneven distributions (see Figure B.3). Channel 1 contains more observations than Channel 2, while region 3 has the highest number of customers. This shows how the dataset is imbalanced across the categorical variables. The pairplot in Figure B.4 shows visible positive relationships among some variables, particularly Milk, Grocery, and Detergents_Paper. This is also confirmed by the heatmap correlation that is presented in Figure B.5 with the strongest correlation observed between Grocery and Detergents_Paper, followed by Grocery and Milk, and Milk and Detergents_Paper. In conclusion, these results indicate how the dataset contains

skewed distributions, exhibitions of the outliers, and correlated features. This justifies the use of scaling and using PCA components as inputs for clustering.

Figure B.1

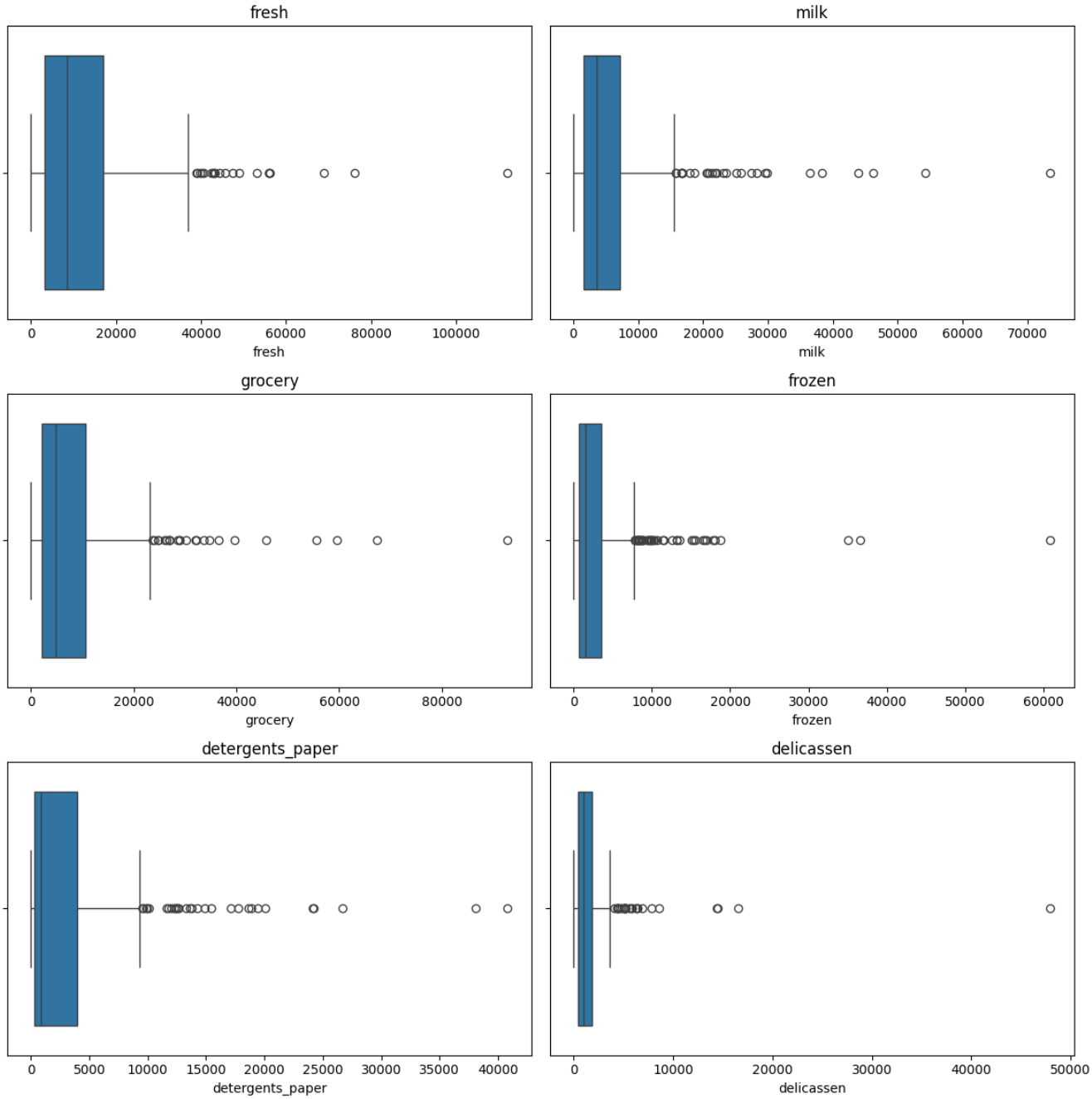
Distribution of Numerical Features



Note. The figure shows histograms for all six numerical features, and the curve is a kernel density estimate (KDE), which approximates the underlying distribution of each feature. All features exhibit right-skewed distributions with long tails, indicating the presence of outliers.

Figure B.2

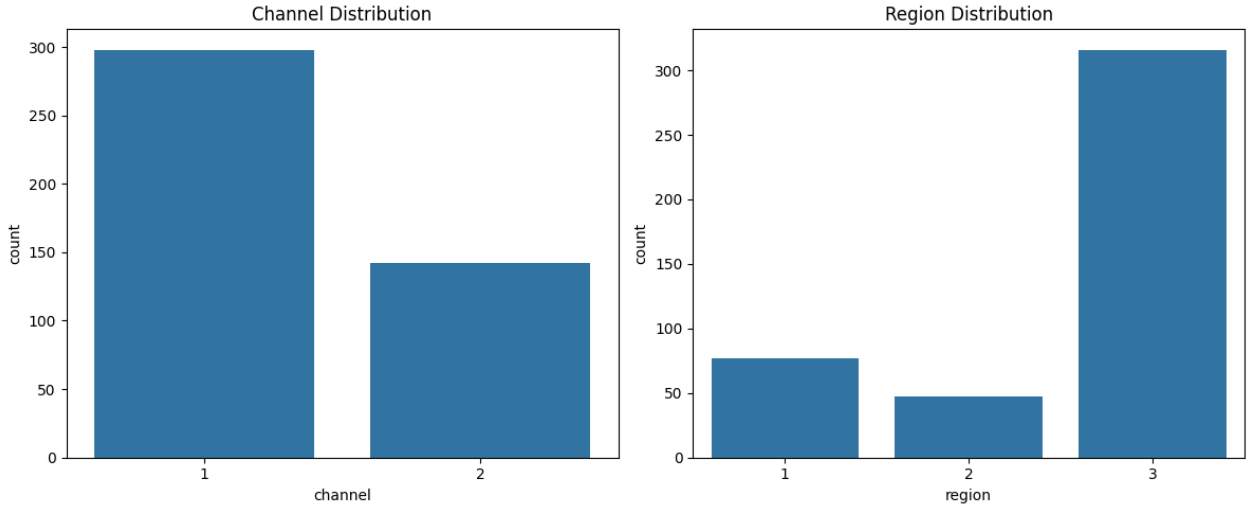
Boxplots of Numerical Features



Note. The figure shows the boxplots for all six numerical features. The presence of outliers beyond the whiskers confirms the right-skewed nature of the data with extreme values.

Figure B.3

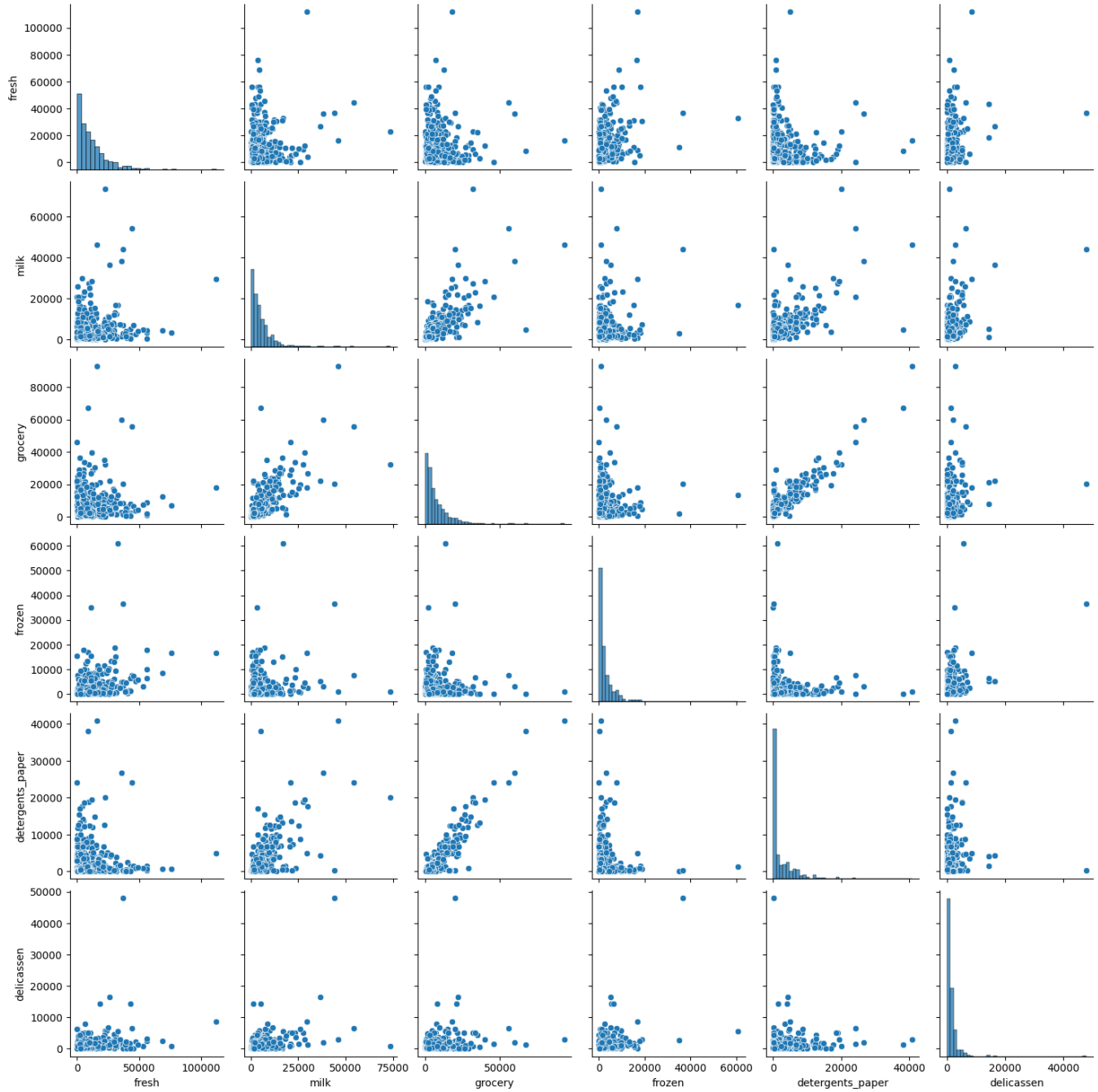
Distribution of Categorical Features



Note. The figure shows the distribution of Channel and Region variables. Both exhibit imbalance, with Channel 1 (Horeca) significantly dominating Channel 2 (Retail), and Region 3 (Other) dominating over Region 1 (Lisbon) and Region 2 (Oporto).

Figure B.4

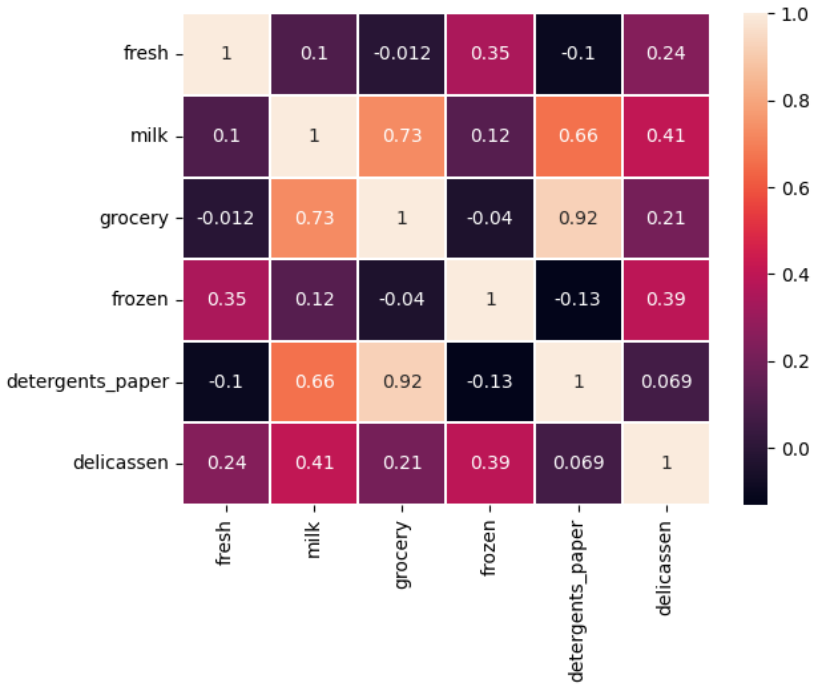
Pairplot of Numerical Features



Note. The figure shows pairwise scatter plots and distributions of the numerical features, revealing strong right-skewness and notable correlations, particularly between Grocery and Detergents_Paper.

Figure B.5

Correlation Heatmap of Numerical Features



Note. The figure shows correlation coefficients between all six numerical features. Strong positive correlations are observed between Grocery and Detergents_Paper (0.92), Grocery and Milk (0.73), and Milk and Detergents_Paper (0.66), suggesting similar purchasing behaviour among these features.

Appendix C: Principal Component Analysis (PCA)

Additional PCA results are reported to support the interpretation. The summary of the explained variance ratio and cumulative variance for each principal component is displayed in Table C.1. The cumulative explained variance plot shows that the first three principal components are sufficient to meet the 80% threshold (see Figure C.1). Figure C.2 shows the individual explained variance plot, and PC1 contributes the largest share of the variation, followed by PC2 and PC3, which add smaller dimensional information. Together, these findings support the idea of using three components for further analysis. Figure C.3 presents the PCA space and how the original variables align with different directions of variation. Instead of showing a clearly separated group, it shows how the customer's purchasing behaviour varies. Average component scores by region and channel are plotted to visualise

how the PCA components are distributed across these variables. In the region heatmap, Region 2 is shown in red for PC1, indicating the highest positive average score, while Regions 1 and 2 appear in blue for PC2 and PC3, reflecting lower average scores on these components (Figure C.4). However, these differences remain relatively modest overall. Channel averages show a clear separation in Figure C.5. Channel 2 has a strongly positive average score on PC1 and a negative value on PC2, while Channel 1 shows the opposite pattern. This supports the interpretation of PC1 as a domestic consumption behavior dimension and PC2 as a culinary or food-preparation purchasing dimension. PC3 varies only slightly across both channels and regions, suggesting that it captures a more specific distinction in customer purchasing behaviour rather than a broad difference across customer groups. Components are labelled based on the loadings and distributions of region and channel, presented in Table C.2.

Table C.1

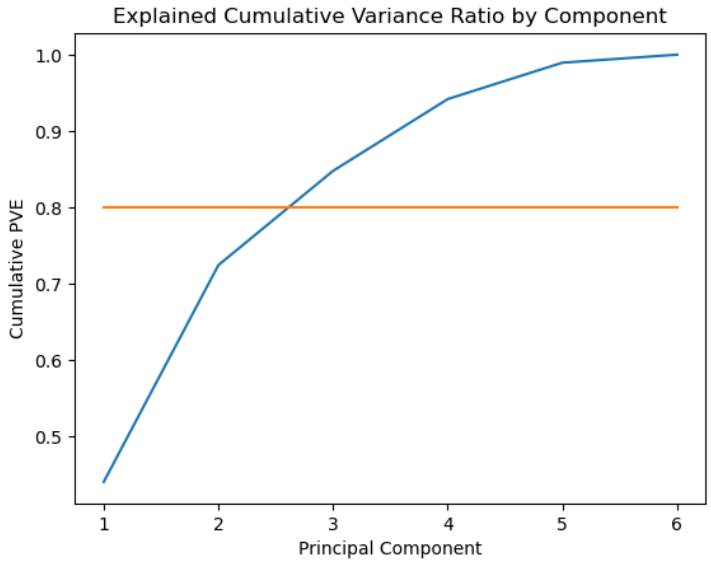
Explained Variance Ratio and Cumulative Variance by Principal Component

Principal Component	Explained Variance Ratio	Cumulative Variance
PCA 1	44.08%	44.08%
PCA 2	28.38%	72.46%
PCA 3	12.33%	84.79%
PCA 4	9.40%	94.19%
PCA 5	4.76%	98.95%
PCA 6	1.05%	100.00%

Note. The table reports the proportion of variance explained by each principal component and the cumulative variance explained across components. The first three components explain 84.79% of the total variance,

Figure C.1

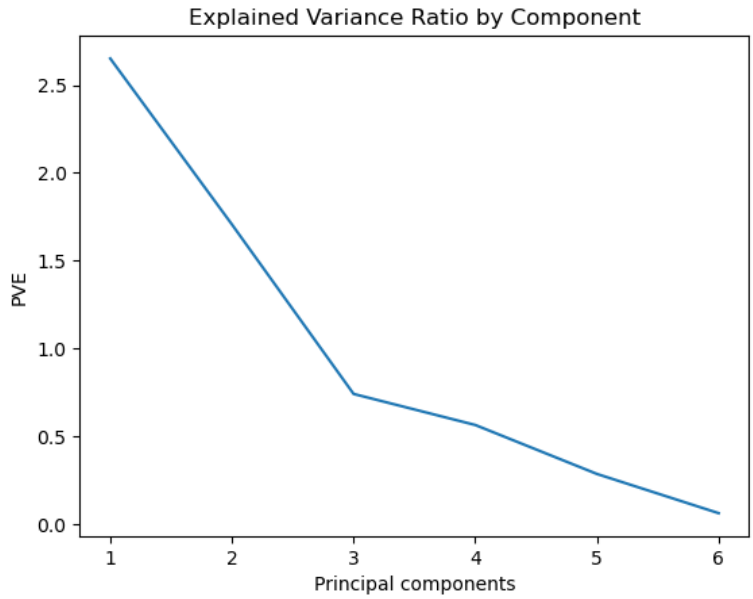
Explained Cumulative Variance Ratio by Principal Component



Note. *The plot shows the cumulative proportion of variance explained by the retained principal components. The horizontal line marks the 80% threshold used to support component selection. To pass the threshold, three components are required.*

Figure C.2

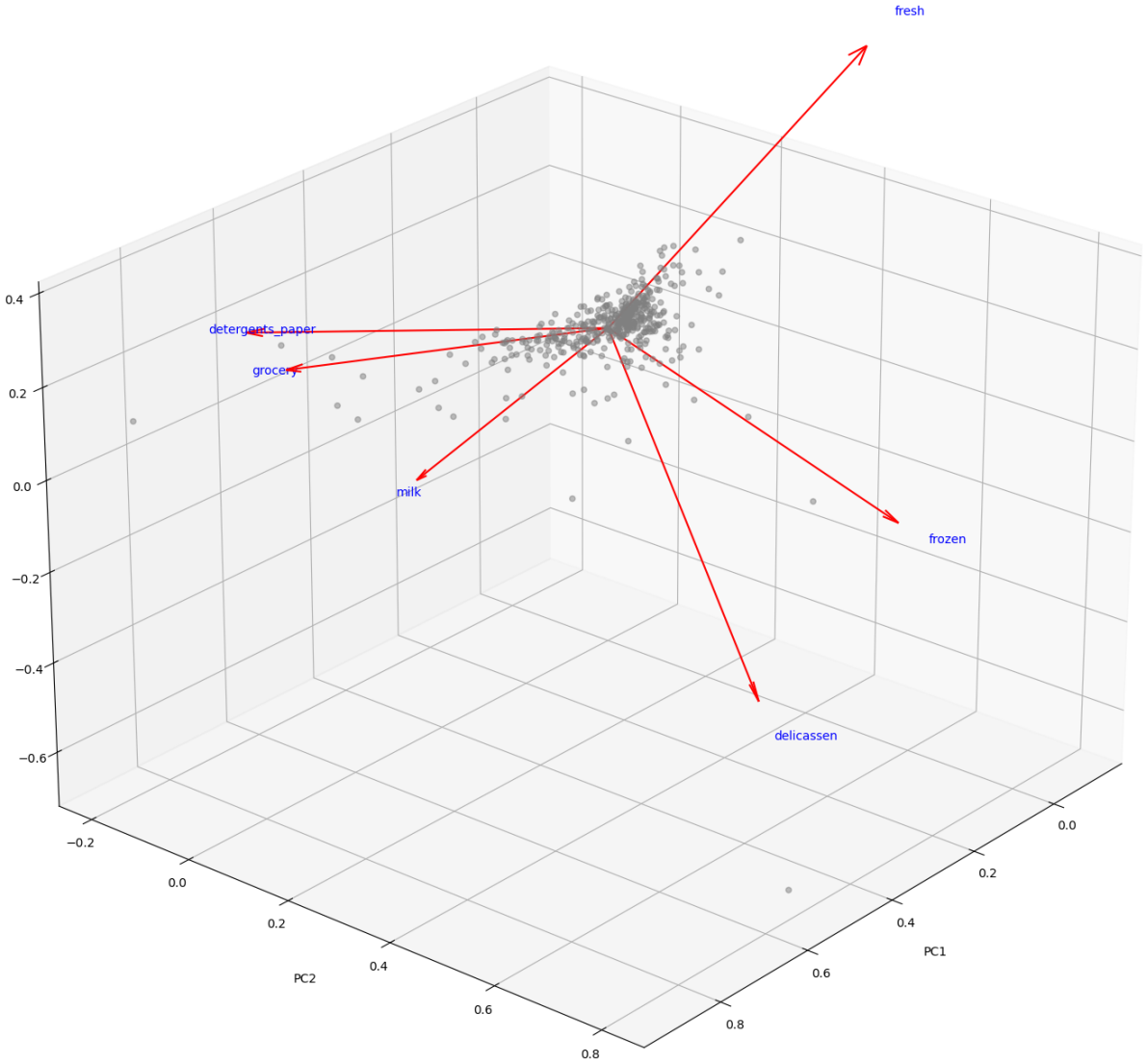
Explained Variance Ratio by Principal Component



Note. The plot presents the variance explained by each individual principal component and highlights the relative contribution of the retained components. This is an elbow method which can reveal the number of components to choose from. From this figure we can see three components is required for PCA analysis.

Figure C.3

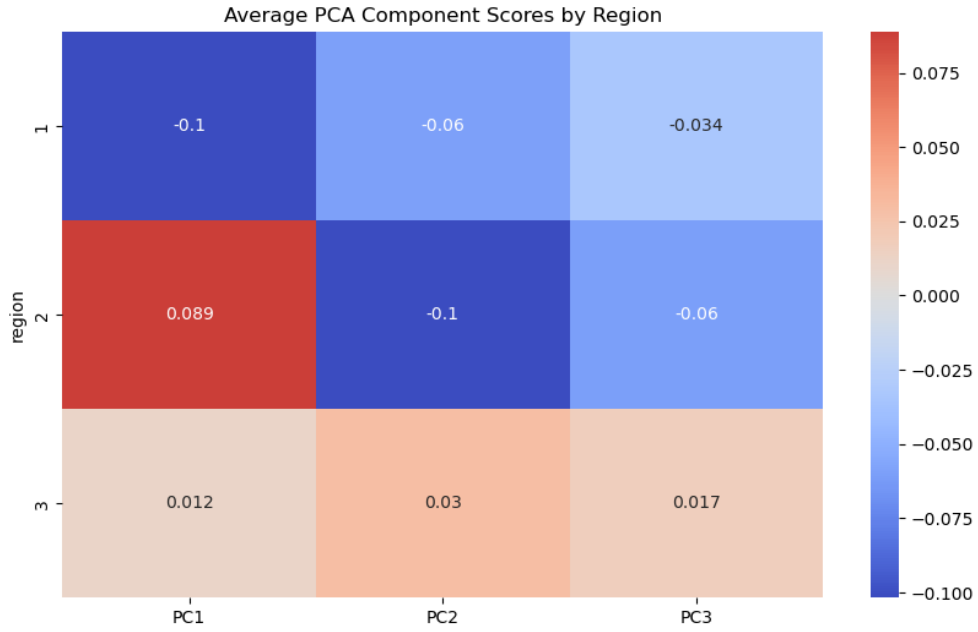
PCA Space for the Retained Principal Components



Note. *The figure presents the observations in the PCA space along with the loading directions of the original variables. The arrows indicate how each variable contributes to the main directions of variation.*

Figure C.4

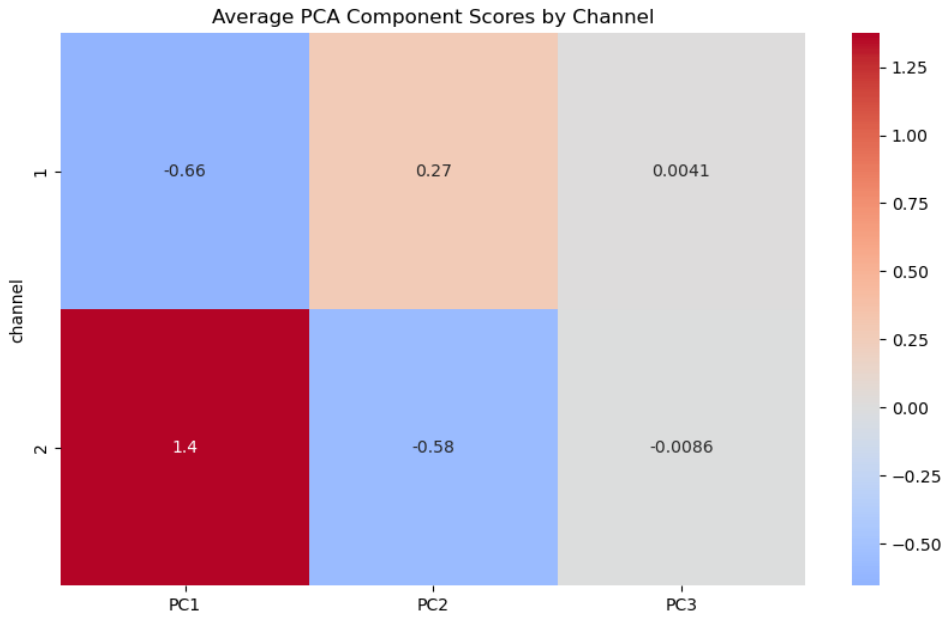
Distribution of Categorical Features



Note. *The heatmap shows average scores for the retained principal components across regions, providing supporting evidence for regional differences in purchasing patterns.*

Figure C.5

Distribution of Categorical Features



Note. The heatmap shows average scores for the retained principal components across customer channels, supporting the interpretation of component differences between retail and HoReCa customers.

Table C.2

Interpretive Labels for the Retained Principal Components

PCA Components	Labels
PCA 1	Domestic Consumption Behaviour
PCA 2	Culinary / Food Preparation Purchase
PCA 3	Fresh Commodity vs Speciality

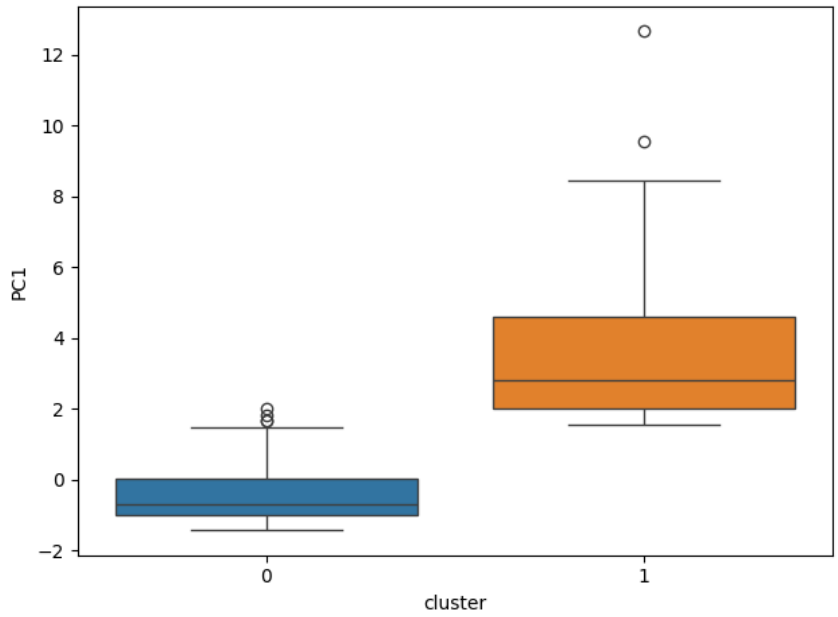
Note. Component labels were assigned based on the dominant loading patterns shown in Table 2 and supported by the channel and region. These labels are interpretive and are intended to summarise the main purchasing pattern captured by each component.

Appendix D: Clustering Analysis

D.1 K-means Clustering Results

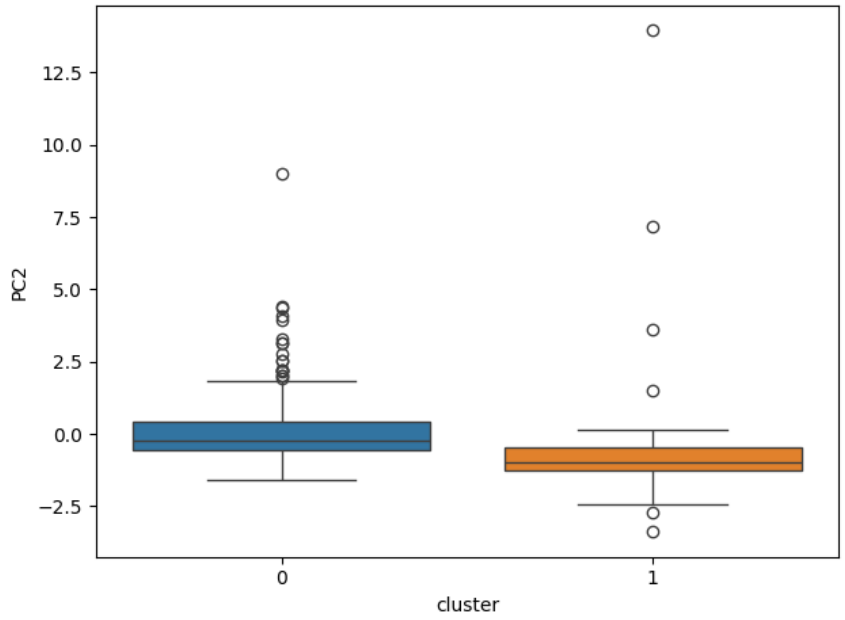
Figure D.1.1

Distribution of PC1 across K-means clusters



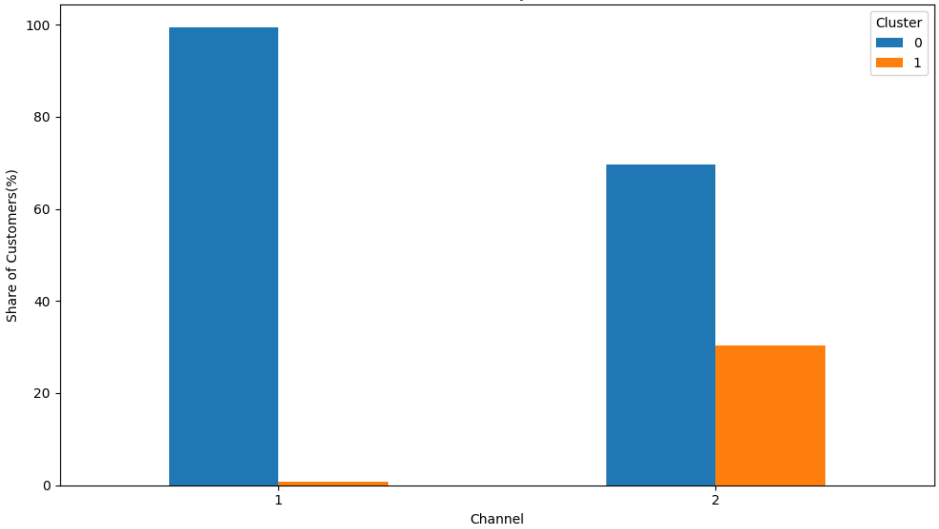
Note. PC1 values for Cluster 1 are higher for Cluster 1 with greater variability, which indicates that Cluster 1 consists of higher domestic consumption customers. On the other hand, Cluster 0 values sit at lower PC1 values with less dispersion.

Figure D.1.2
Distribution of PC2 across K-means clusters



Note. PC2 values for Clusters 1 and 2 are very close to each other, with Cluster 1 sitting slightly at the lower end of the PC2 axis compared to Cluster 2. Similarly, both clusters have very similar and limited variability across PC2, meaning that variables are not well-explained by PC2 and are mainly explained by PC1.

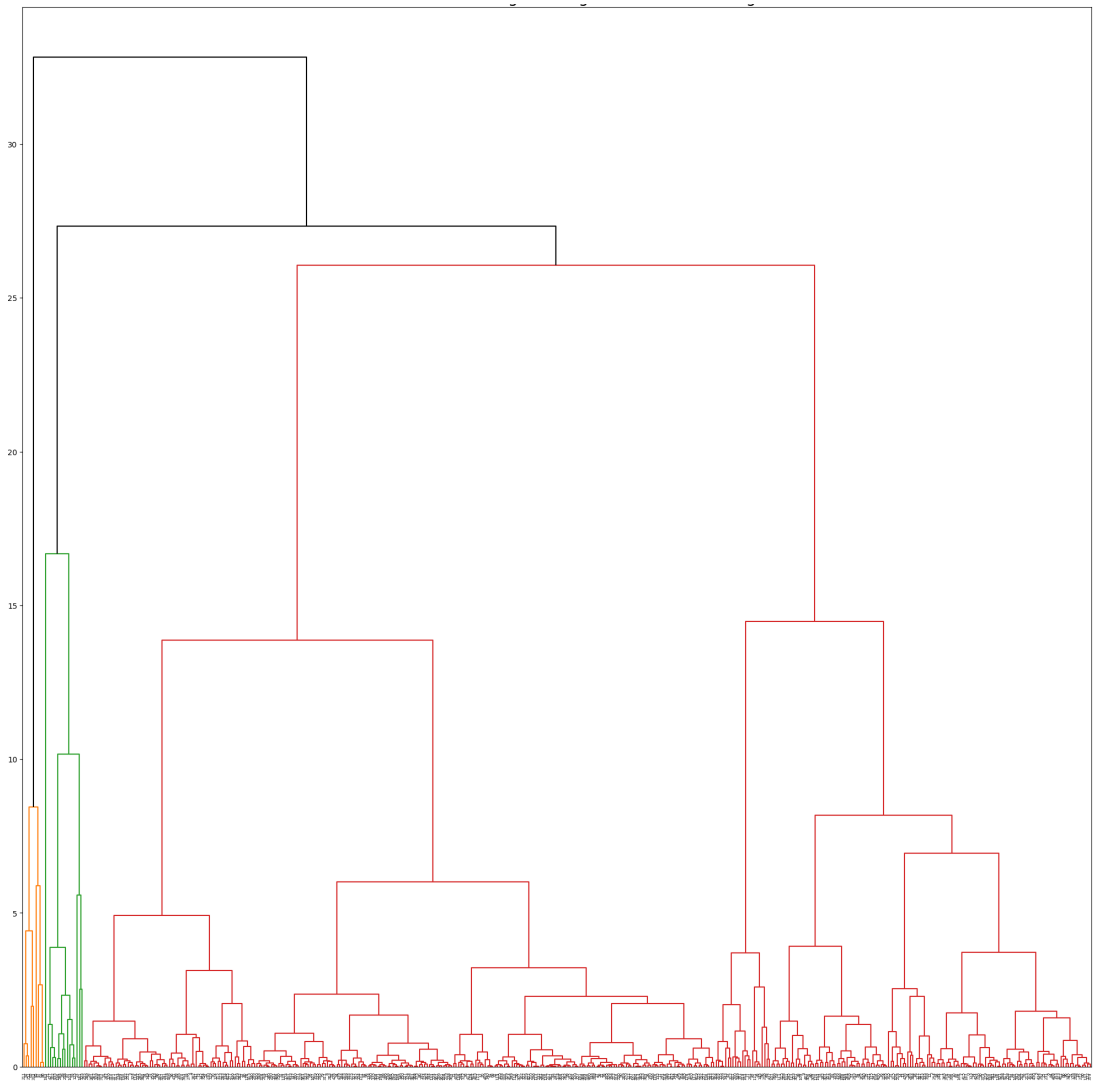
Figure D.1.3
Distribution of Customer Clusters Across Channels



Note. Cluster 0 dominates both Channel 1 (Horeca) and Channel 2 (Retail), which indicates that, regardless of the channel, most customers belong to the mainstream cluster, especially in Channel 1. Cluster 1 appears only marginally in Channel 1 while being more prominent in Channel 2, indicating a channel dominance.

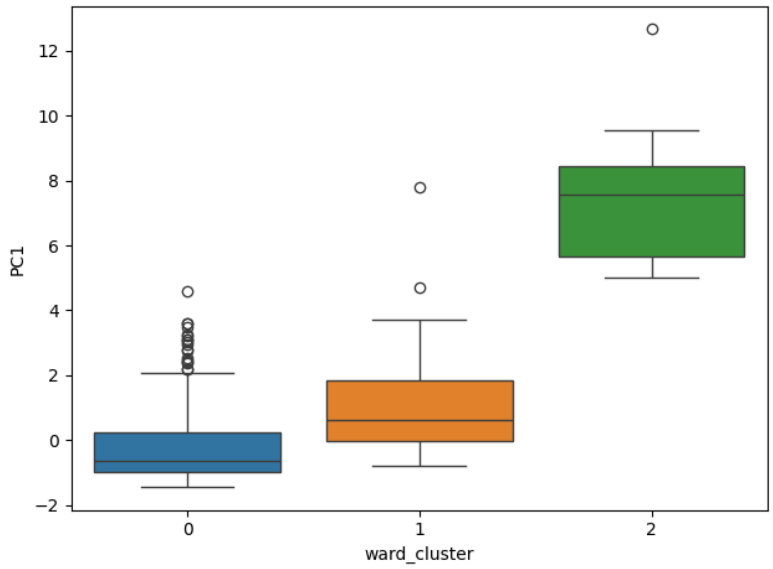
D.2 Hierarchical Clustering Ward Linkage Results

Figure D.2.1
Hierarchical clustering with Ward linkage dendrogram



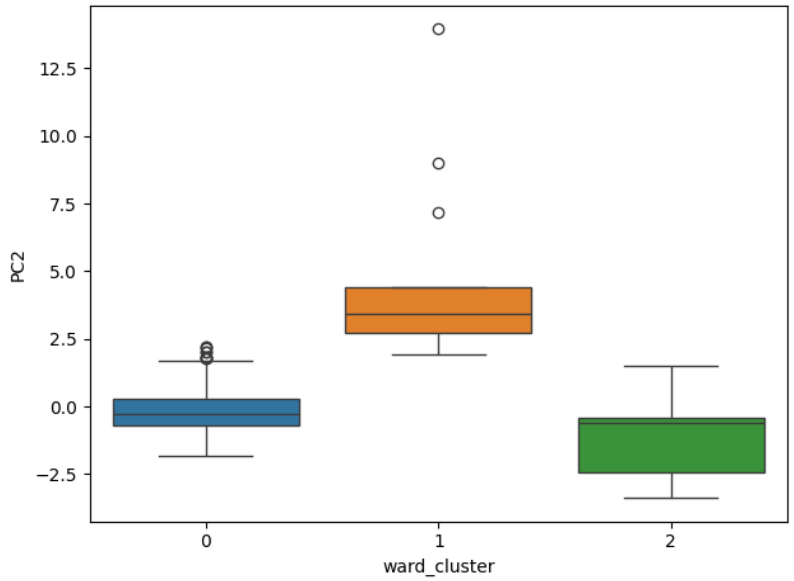
Note. The dendrogram is composed of two axes. While the vertical axis displays the merge distance height of clusters, the horizontal axis represents each observation. The figure suggests that there are 3 distinct clusters with different sizes at the optimal cut-off height of 27.

Figure D.2.2
Distribution of PC1 across Hierarchical clustering with Ward linkage *clusters*



Note. PC1 values for Cluster 1 are highest for Cluster 2, with the greatest variability, which indicates that Cluster 2 consists of higher domestic consumption customers. On the other hand, Cluster 0 values sit at lower PC1 values with less dispersion.

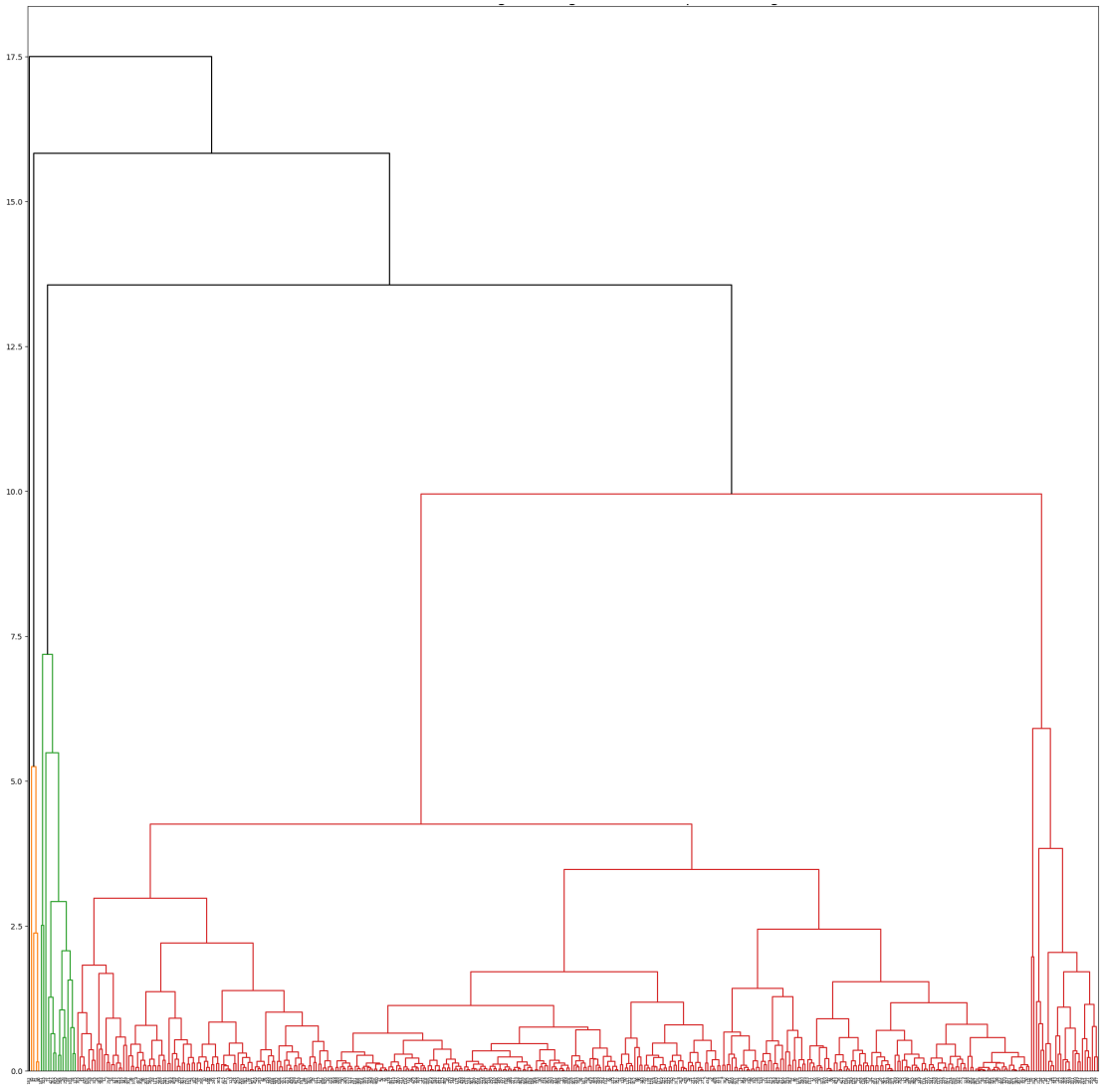
Figure D.2.3
Distribution of PC2 across Hierarchical clustering with Ward linkage *clusters*



Note. PC2 values for Clusters 1, 2 and 3 also differ, with Cluster 1 sitting at the higher end of the PC2 axis compared to the others. This indicates that Cluster 1 customers are highly culinary/ food prep spending oriented, while Cluster 2 is completely opposite. Similarly, all clusters have different variability across PC2, meaning that variables are well-explained by PC2

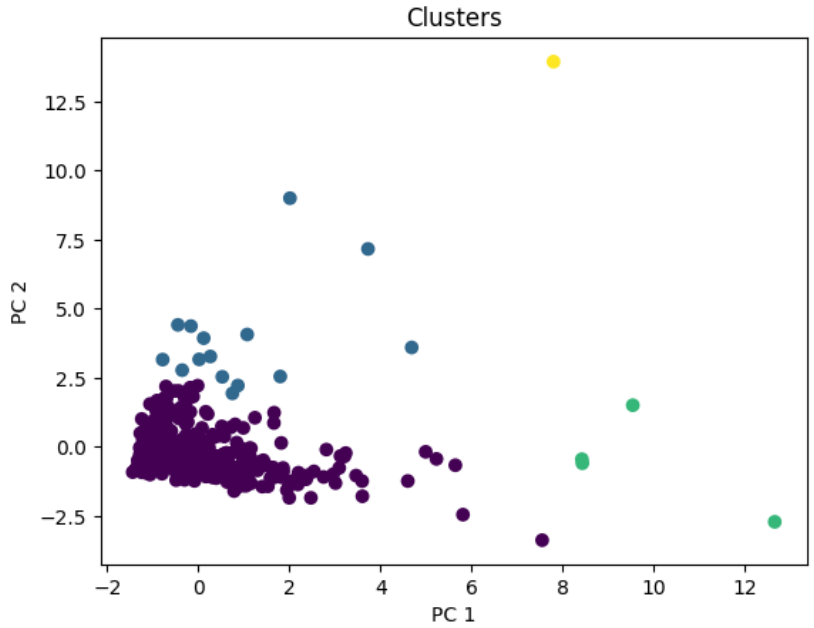
D.3 Hierarchical Clustering Complete Linkage Results

Figure D.3.1
Hierarchical clustering with Complete linkage dendrogram



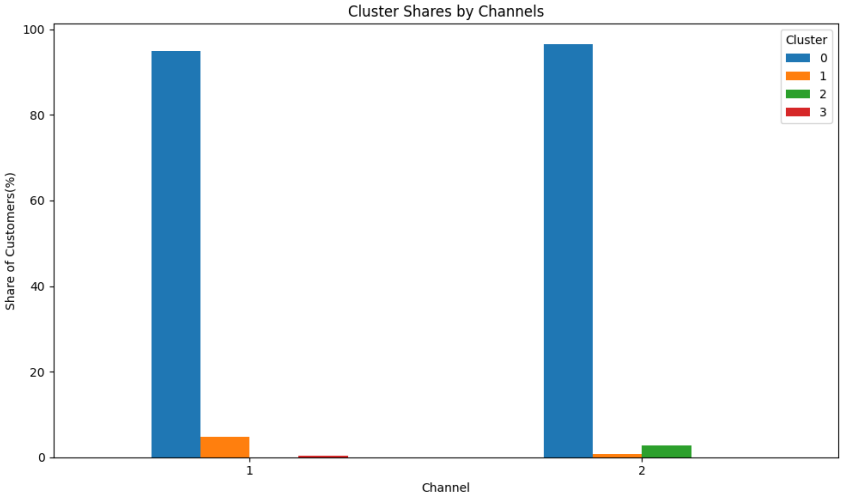
Note. The dendrogram is composed of two axes. While the vertical axis displays the merge distance height of clusters, the horizontal axis represents each observation. The figure suggests that there are 4 distinct clusters with a notable singleton cluster at the optimal cut-off height of 12.5.

Figure D.3.2
Complete Linkage Clusters in PCA-Transformed Feature Space



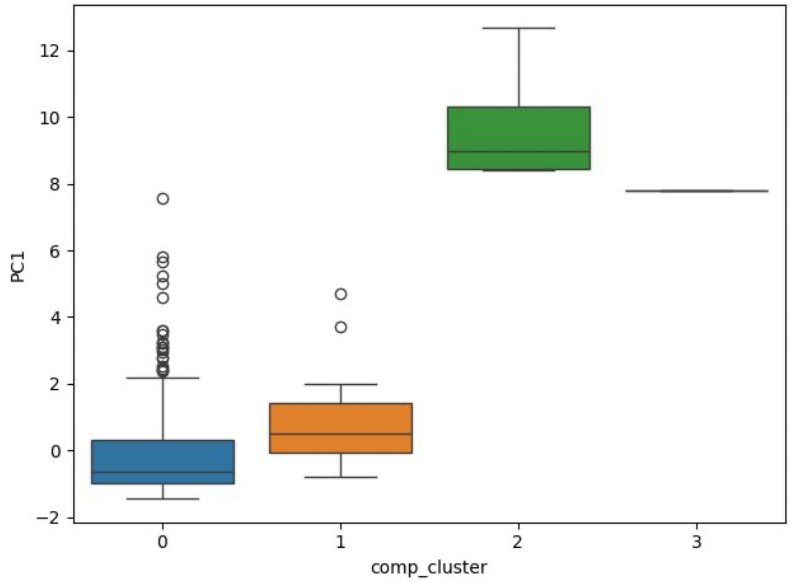
Note: The majority of observations are in Cluster 0, purple. Clusters mainly differ across the PC1 axis. Clusters show variation between clusters in domestic consumption.

Figure D.3.3
Distribution of Customer Clusters Across Channels



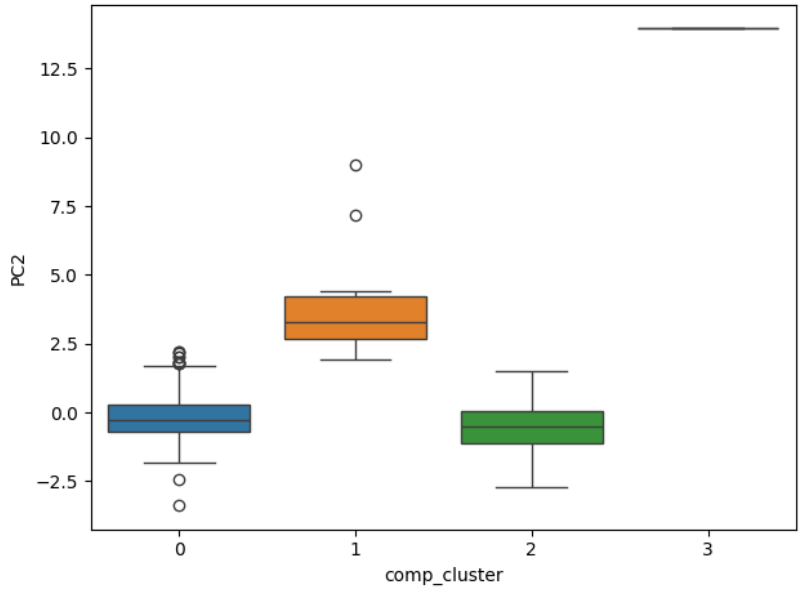
Note. Cluster 0 dominates both Channel 1 (Horeca) and Channel 2 (Retail), which indicates that, regardless of the channel, most customers belong to the mainstream cluster. Cluster 1 appears only marginally in both channels but more in Channel 1, indicating a specialised consumption pattern rather than channel domination while Channel 2 indicates channel dominance. And Cluster 3 is barely represented in Channel 1.

Figure D.3.4
Distribution of PC1 across Hierarchical clustering with Complete linkage *clusters*



Note. PC1 values for Cluster 2 are the highest for Cluster 2, but displaying extreme values, which indicates that Cluster 2 might be representing outliers or extreme buyers. On the other hand, Cluster 0 values sit at lower PC1 values, characterised by less domestic consumption behaviour.

Figure D.3.5
Distribution of PC2 across Hierarchical clustering with Complete linkage *clusters*



Note. PC2 values for Clusters 0 and 2 are very close to each other, with Cluster 2 sitting slightly at the lower end of the PC2 axis compared to Cluster 0. Similarly, most clusters exhibit comparable variability across PC2, suggesting that PC2 does not strongly differentiate between clusters. PC2 represents extreme values or outliers in this case.

Appendix E: LLM Prompts and Outputs

This appendix presents the full prompts and corresponding outputs generated by Google Gemini Pro.

Prompt 1:

> *The attached data set refers to clients of a wholesale distributor. It includes the annual spending in monetary units (m.u.) on diverse product categories such as milk, groceries, and fresh products. It also includes identifier information such as channel and region. As a Machine Learning Master's student, your task is to first conduct a principal component analysis on the dataset, followed by clustering the customers (both through K-means and Hierarchical Clustering Methods) based on their purchase behaviours using the PCA-transformed data from the wholesale provider. Give me the full Python pipeline to be able to conduct this full customer segmentation task.*

Output 1:

This task involves typical unsupervised learning workflows: dimensionality reduction followed by clustering. Using Principal Component Analysis (PCA) before clustering helps in reducing noise and multicollinearity, while also allowing for easier visualization of the customer segments.

Python Pipeline for Customer Segmentation

The following pipeline handles data preprocessing, PCA, and the comparison between K-means and Hierarchical clustering.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from sklearn.cluster import KMeans, AgglomerativeClustering
from scipy.cluster.hierarchy import dendrogram, linkage

# 1. Load Dataset
df = pd.read_csv('Wholesale customers data.csv')
```

```
# 2. Preprocessing
features = ['Fresh', 'Milk', 'Grocery', 'Frozen', 'Detergents_Paper',
'Delicassen']
X = df[features]
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# 3. PCA
pca_full = PCA()
pca_full.fit(X_scaled)
plt.figure(figsize=(8, 4))
plt.plot(range(1, 7), pca_full.explained_variance_ratio_.cumsum(),
marker='o')
plt.title('Cumulative Explained Variance by PCA Components')
plt.xlabel('Number of Components')
plt.ylabel('Variance Explained')
plt.grid(True)
plt.show()

pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)

# 4. K-means Clustering
wcss = []
for i in range(1, 11):
    km = KMeans(n_clusters=i, init='k-means++', random_state=42, n_init=10)
    km.fit(X_pca)
    wcss.append(km.inertia_)

plt.figure(figsize=(8, 4))
plt.plot(range(1, 11), wcss, marker='o', color='red')
plt.title('K-means Elbow Method')
plt.show()

kmeans = KMeans(n_clusters=3, init='k-means++', random_state=42, n_init=10)
df['KMeans_Cluster'] = kmeans.fit_predict(X_pca)
```

```
# 5. Hierarchical Clustering
plt.figure(figsize=(10, 6))
linkage_matrix = linkage(X_pca, method='ward')
dendrogram(linkage_matrix)
plt.title('Hierarchical Clustering Dendrogram')
plt.ylabel('Euclidean Distances')
plt.show()

agg = AgglomerativeClustering(n_clusters=3, metric='euclidean',
linkage='ward')
df['Hierarchical_Cluster'] = agg.fit_predict(X_pca)

# 6
. Visualization
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(16, 6))
ax1.scatter(X_pca[:, 0], X_pca[:, 1], c=df['KMeans_Cluster'], cmap='viridis',
edgecolors='k')
ax1.set_title('K-means Segmentation (PCA Space)')
ax2.scatter(X_pca[:, 0], X_pca[:, 1], c=df['Hierarchical_Cluster'],
cmap='plasma', edgecolors='k')
ax2.set_title('Hierarchical Segmentation (PCA Space)')
plt.tight_layout()
plt.show()
```

Analysis Overview

1. Preprocessing: Standardizing the features ensures that high-magnitude categories (like Fresh) do not dominate categories with lower spending (like Delicassen) during distance calculations.
2. PCA: The first two principal components typically capture over of the total variance in this dataset. These components often represent "Bulk Grocery Spending" vs "Fresh/Frozen Food Focus."
3. K-means: This method partitions the PCA-reduced data into spherical clusters. It is computationally efficient but assumes clusters are convex.
4. Hierarchical Clustering: Using the Ward linkage method minimizes the variance within clusters. The resulting Dendrogram provides a multi-level perspective on how individual customers are grouped, allowing for more flexible selection of the number of segments.

Key Differences

- K-means is faster and better for large datasets but requires the number of clusters to be pre-specified.
- Hierarchical Clustering is more informative for small-to-medium datasets because it shows the relationship between all clusters, though it is more computationally expensive.

Prompt 2:

> *which one is best model for clustering based on your analysis*

Output 2:

Based on the analysis of the wholesale customer data, the Hierarchical Clustering (Agglomerative) model performed slightly better than K-means for this specific dataset and PCA transformation.

1. Performance Comparison (Silhouette Score)

The Silhouette Score is a standard metric used to evaluate how well each data point fits into its assigned cluster compared to others (ranging from -1 to 1, where higher is better).

- Hierarchical Clustering (Ward's Linkage): ~0.67
- K-means Clustering: ~0.63

The higher score for Hierarchical Clustering suggests that it produced segments that are more distinct and have higher internal cohesion.

2. Qualitative Strengths

While K-means is highly efficient for larger datasets, Hierarchical Clustering offers several advantages for this specific wholesale distributor dataset (which has 440 records):

- Dendrogram Interpretability: The dendrogram (tree-like diagram) allows us to see exactly how different customers are grouped at various levels of granularity. This is invaluable for a distributor who might want to understand "sub-segments" (e.g., distinguishing between different types of retail outlets or restaurants).
- No Initial "K" Required: Unlike K-means, which requires you to guess the number of clusters (often via the Elbow Method), Hierarchical clustering allows you to observe the natural hierarchy of the data before deciding where to "cut the tree."
- Robustness to Cluster Shapes: K-means assumes clusters are spherical and roughly equal in size. In wholesale spending, where some customers (like large supermarket chains) are massive outliers compared to small cafes, Hierarchical clustering (specifically using Ward's method) is often more robust at handling these variations in cluster density.

Summary of Customer Segments

Regardless of the model chosen, the PCA-transformed data consistently revealed three primary segments:

1. High-Volume Retailers: Large spenders on Grocery and Detergents/Paper (typically "Channel 2" customers).
2. Fresh/Food Service Focus: Customers with high spending on Fresh and Frozen products (typically "Channel 1" hotels/restaurants).
3. Low-Volume/Niche Buyers: Smaller, more frequent buyers who have balanced spending but lower total annual volume.

Conclusion: For this task, Hierarchical Clustering is the preferred model because it provides a more accurate mathematical separation (higher silhouette score) and offers superior diagnostic visualization through the dendrogram.

Prompt 3:

How did you choose cluster numbers in both K-Means and Hierarchical Clustering?

Output 3:

To determine the most appropriate number of clusters, I used two distinct mathematical and visual heuristics common in machine learning: the Elbow Method for K-Means and Dendrogram Analysis for Hierarchical Clustering.

1. K-Means: The Elbow Method

K-Means aims to minimize the Within-Cluster Sum of Squares (WCSS), which is the sum of squared distances between each data point and its assigned cluster centroid.

- The Logic: As you increase the number of clusters (), the WCSS will naturally decrease because the centroids get closer to the data points.
- The "Elbow": We look for the point where the rate of decrease shifts from rapid to slow. This creates an "elbow" shape in the plot.
- Application: In our analysis, the drop in WCSS was significant from to , but the gain in "compactness" became much smaller after . Thus, was chosen as the point of diminishing returns.

2. Hierarchical Clustering: Dendrogram Analysis

Hierarchical clustering doesn't start with a fixed number of clusters. Instead, it builds a tree-like structure (a dendrogram) by merging pairs of clusters based on their proximity.

- The Logic: The vertical axis represents the Euclidean distance (dissimilarity) between clusters. The longer the vertical lines, the more different the clusters are.
- The "Horizontal Cut": To choose the number of clusters, we look for the largest vertical distance that does not intersect any horizontal merge line. This represents the most "natural" separation in the data.
- Application: By visualizing the dendrogram for the wholesale data, cutting the tree at a distance that yields 3 clusters provided the most balanced and distinct groupings. It separated the "big spenders" from the "specialized food service" and "small scale" clients effectively.

3. Validation: Silhouette Score

To confirm these visual choices were mathematically sound, I cross-validated both methods using the Silhouette Score.

The Silhouette Score measures how similar an object is to its own cluster (cohesion) compared to other clusters (separation). The score for k=3 was higher than for k=2 or k=4, confirming that 3 clusters was the "sweet spot" where customers were most clearly differentiated by their purchasing behavior.

How do these segments align with your expectations for the wholesale market? For instance, does a three-segment split (Retailers, Food Service, and Small-scale) fit your business use case?

Prompt 4:

> *Why did you use Ward linkage on hierarchical clustering?*

Output 4:

I chose Ward's Linkage (also known as the minimum variance method) because it is specifically designed to create compact, even-sized clusters, which aligns perfectly with the goals of customer segmentation.

In a business context like wholesale distribution, we aren't just looking for "connections" between data points; we are looking to partition the customer base into distinct, manageable groups. Here is why Ward's is the industry standard for this task:

1. Minimizing Within-Cluster Variance

While other methods (like Single or Complete linkage) focus on the distance between the *closest* or *farthest* points of two clusters, Ward's linkage focuses on the internal cohesion. It merges clusters that result in the smallest increase in total within-cluster sum of squares (WCSS).

Mathematically, when merging two clusters and , Ward's looks at the increase in variance:

$$\Delta(A, B) = \frac{n_A n_B}{n_A + n_B} \|\bar{m}_A - \bar{m}_B\|^2$$

(Where is the number of points and is the centroid.)

2. Handling Outliers and "Chaining"

Other linkage methods often suffer from chaining, where clusters become long and "straggly" because they keep adding nearby points one by one.

- Single Linkage can lead to one giant cluster and several tiny ones.
- Ward's Linkage actively resists this by preferring to merge smaller clusters of similar density, resulting in more "spherical" and distinct groups.

3. Consistency with K-Means

Since the objective was to compare K-Means and Hierarchical methods, Ward's linkage is the most logical choice. Both K-Means and Ward's Hierarchical clustering share the same objective function: minimizing the sum of squared errors. By using Ward's, we ensure that the differences we see between the two models are due to the *structure* of the algorithms (partitioning vs. hierarchical) rather than entirely different mathematical goals.

4. Business Interpretability

For a wholesale distributor, a cluster that is "compact" means the customers within that group have very similar spending habits.

- Ward's produces groups where the "average" customer of the group actually represents the individuals well.
- This makes it much easier for a marketing team to design a single strategy (e.g., a "Fresh Food" promotion) that appeals to the entire cluster.

Would you like to see how the cluster characteristics change if we were to use a different linkage method, like Complete Linkage, which focuses on the maximum distance between points instead?